

# ML-DL-Ops : Assignment 5

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## Introduction

This assignment is about parameter-efficient fine-tuning using LoRA on Vision Transformers and robustness of neural networks under adversarial attacks using FGSM, PGD, and BIM. The goal is to analyze performance, efficiency, and robustness trade-offs.

## Q1: ViT Fine-tuning with LoRA

### Overview

In this task, a pre-trained Vision Transformer (ViT-Small) model was fine-tuned on the CIFAR-100 dataset using two approaches:

- **Baseline:** Only the classification head was fine-tuned while keeping the backbone frozen.
- **LoRA-based Fine-tuning:** Low-Rank Adaptation (LoRA) was applied to the attention layers (Query, Key, Value), while keeping the classification head trainable.

LoRA introduces trainable low-rank matrices into the frozen weights, allowing efficient adaptation with significantly fewer parameters compared to full fine-tuning. The experiments were conducted for different hyperparameter combinations:

- Rank  $r \in \{2, 4, 8\}$
- Alpha  $\alpha \in \{2, 4, 8\}$
- Dropout = 0.1

All combinations of rank and alpha were evaluated, and the best-performing configuration was found to be:

**Rank = 8, Alpha = 8, Dropout = 0.1**

This configuration achieved the highest validation and test performance, demonstrating that higher-rank adapters provide better representational capacity while still being parameter-efficient.

### Training Table (Best Model: Rank=8, Alpha=8)

| Epoch | Train Loss | Val Loss | Train Acc | Val Acc |
|-------|------------|----------|-----------|---------|
| 1     | 0.6572     | 0.3703   | 82.65     | 88.52   |
| 2     | 0.2705     | 0.3563   | 91.46     | 89.06   |
| 3     | 0.1858     | 0.3558   | 94.04     | 89.55   |
| 4     | 0.1259     | 0.3685   | 96.01     | 89.67   |
| 5     | 0.0849     | 0.3798   | 97.50     | 89.63   |
| 6     | 0.0553     | 0.3830   | 98.51     | 89.96   |
| 7     | 0.0371     | 0.3862   | 99.22     | 89.92   |
| 8     | 0.0270     | 0.3893   | 99.49     | 89.92   |
| 9     | 0.0211     | 0.3933   | 99.72     | 90.04   |
| 10    | 0.0187     | 0.3941   | 99.79     | 90.01   |

### Testing Table (ALL COMBINATIONS)

| LoRA | Rank | Alpha | Dropout | Test Acc     | Params |
|------|------|-------|---------|--------------|--------|
| No   | -    | -     | -       | 79.43        | 38K    |
| Yes  | 2    | 2     | 0.1     | 89.63        | 75K    |
| Yes  | 2    | 4     | 0.1     | 89.76        | 75K    |
| Yes  | 2    | 8     | 0.1     | 89.76        | 75K    |
| Yes  | 4    | 2     | 0.1     | 89.96        | 112K   |
| Yes  | 4    | 4     | 0.1     | 89.95        | 112K   |
| Yes  | 4    | 8     | 0.1     | 89.71        | 112K   |
| Yes  | 8    | 2     | 0.1     | 89.99        | 185K   |
| Yes  | 8    | 4     | 0.1     | 90.06        | 185K   |
| Yes  | 8    | 8     | 0.1     | <b>90.15</b> | 185K   |

### Observations

- LoRA improves accuracy from 79.43%  $\rightarrow$  90.15% - Higher rank  $\rightarrow$  better learning capacity - Best config: Rank=8, Alpha=8 - Huge parameter reduction vs full fine-tuning

### Q2: Adversarial Attacks

This section evaluates the robustness of a ResNet-18 model on CIFAR-10 against adversarial attacks, specifically comparing FGSM implemented from scratch and using IBM ART. Additionally, adversarial detection models using ResNet-34 are trained to distinguish between clean and perturbed images generated via FGSM, PGD, and BIM attacks.

## FGSM Comparison

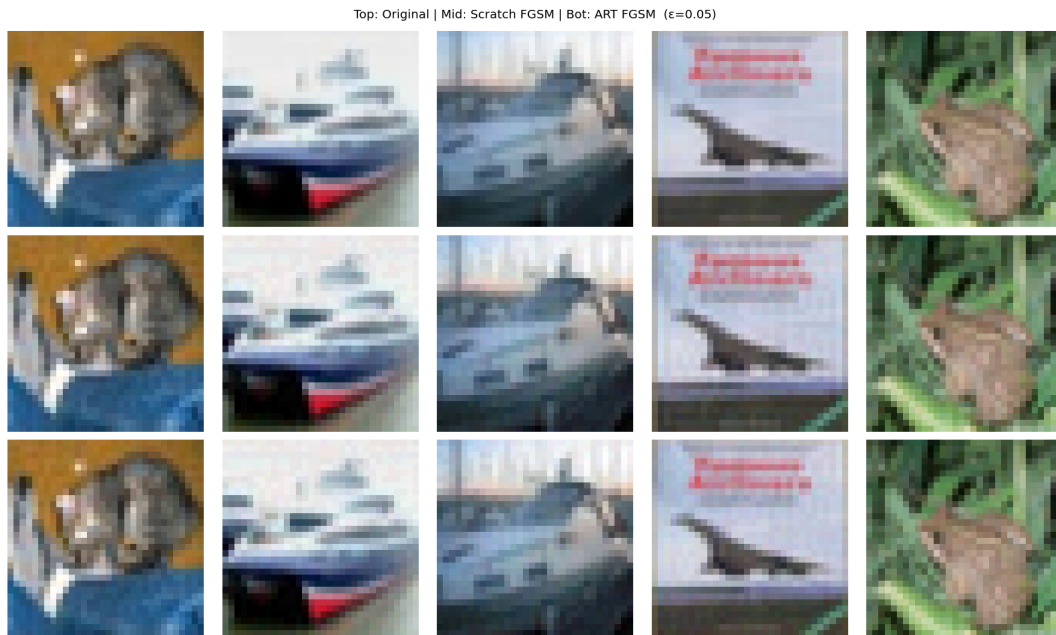


Figure 1: Top: Original, Middle: FGSM Scratch, Bottom: FGSM ART

**Explanation:** Even though images look similar, small perturbations drastically change model predictions. ART implementation produces slightly smoother perturbations compared to scratch.

## Multiple Attack Samples

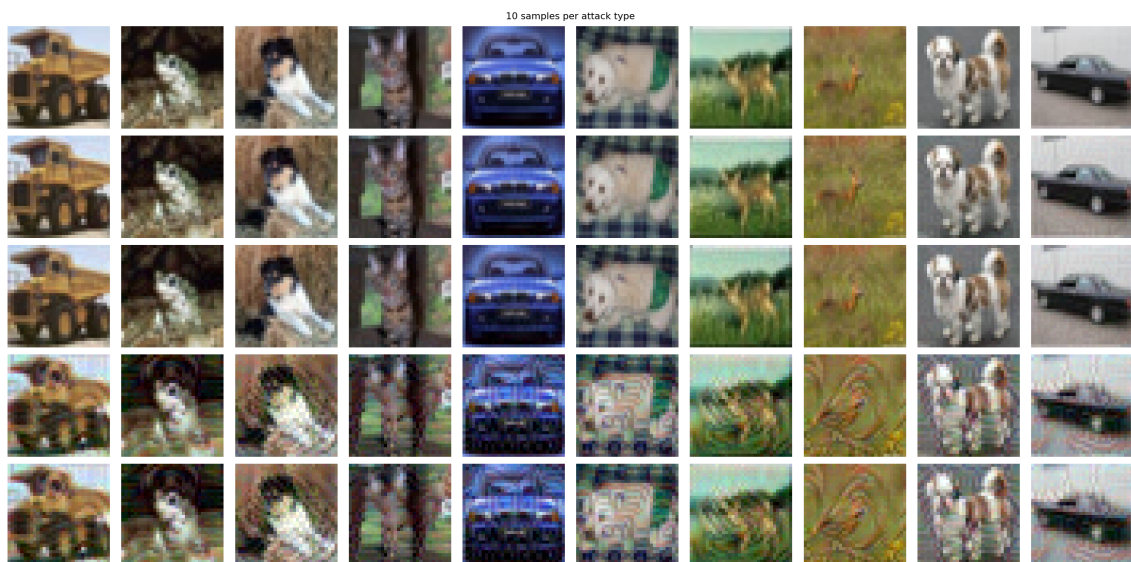


Figure 2: Samples across FGSM, PGD, BIM

**Explanation:** PGD and BIM attacks introduce structured noise patterns. These are harder for models to classify correctly but easier for detectors to identify.

## FGSM Accuracy Comparison

| Scenario     | Accuracy |
|--------------|----------|
| Clean        | 73.64    |
| FGSM Scratch | 56.06    |
| FGSM ART     | 65.48    |

**Observation:** Accuracy drops significantly under attack. Scratch FGSM is stronger due to improper normalization handling.

## Perturbation vs Accuracy

| Epsilon | Accuracy |
|---------|----------|
| 0.01    | 65.48    |
| 0.05    | 28.38    |
| 0.1     | 18.13    |

**Observation:** Higher perturbation  $\rightarrow$  lower accuracy.

## Adversarial Detection

| Attack | Detection Accuracy |
|--------|--------------------|
| PGD    | 99.91              |
| BIM    | 99.94              |

**Observation:** Detection accuracy is very high because adversarial noise follows identifiable patterns.

## Conclusion

LoRA enables efficient fine-tuning with minimal parameters while achieving high accuracy. Adversarial attacks significantly degrade model performance, but detection models effectively identify adversarial samples. This highlights the importance of both robustness and efficiency in deep learning systems.